

Security+ 3.1 · Network Infrastructure Concepts

Master CyberSec — Official CompTIA Security+ (SY0-701) Comprehensive Study Guide · **Lesson 3.1** —
Network Infrastructure Concepts (6:56) · [Official YouTube Video ↗](#) · [Professor Messer Study Hub ↗](#)

Executive Lesson Summary: Network segmentation techniques: VLANs logically segment networks on the same physical hardware; SDN (Software-Defined Networking) separates the control plane from the data plane, enabling programmable and dynamic network rules; microsegmentation isolates individual workloads, implementing Zero Trust at the network layer; VPC (Virtual Private Cloud) provides an isolated network within a cloud provider.

High-Yield Core Concepts Tested on Exam

- SDN: centralized control plane, programmable and dynamic rules
- Microsegmentation: isolates each workload — Zero Trust applied to network
- VPC: isolated network within a cloud provider
- VLAN: logical segmentation on the same physical switch

Exam Tips & Classic Scenarios

- 'centralized control plane managing network routing' = SDN
- 'isolates individual workloads' = microsegmentation
- 'logical network segmentation on same hardware' = VLAN

Typical Exam Scenario: A cloud network is configured so that test workloads cannot communicate with production workloads. This is an example of microsegmentation.

Beyond the Video: Official CompTIA Exam Concepts, Protocols & Traps (Lesson 3.1 — Network Infrastructure Concepts)

Why is this advanced coverage required for SY0-701? Video overviews provide foundational familiarity, but CompTIA tests practical engineering judgment and requires deep familiarity with the following protocols, acronyms, and operational nuances in this domain:

- **WAF:** Web Application Firewall: Layer 7 security inspection appliance/service that inspects and filters HTTP/HTTPS payloads against OWASP Top 10 exploits such as SQL Injection, XSS, and CSRF.

Classic CompTIA Traps & Question Framing:

- Avoid rote memorization: CompTIA exam prompts regularly adjust technical constraints requiring you to defend why one valid security control is the *MOST appropriate* or *FIRST step*.
- Scrutinize question qualifiers: 'without additional licensing costs' favors native OS tools; 'granular administrative command accounting' favors TACACS+ over RADIUS; 'non-repudiation' mandates asymmetric private key signing.

Practice Questions with In-Depth Solutions (10 Questions)

Detailed answer key with step-by-step technical rationale and comprehensive distractor breakdown provided below. Attempt answering independently first.

1. Your isolated private network within the cloud, where you define subnets and routing:

- A)** VPC (Virtual Private Cloud)
- B)** CDN
- C)** WAF
- D)** NAT gateway

2. What does VPC stand for and what is its primary function?

- A)** Virtual Private Cloud - isolated network in the cloud
- B)** Virtual Public Cloud - shared network
- C)** Virtual Computing - elastic processing
- D)** Volume Persistent Cloud - storage

3. Using MULTIPLE layers of defense (firewall + antivirus + MFA + training) so as not to depend on just one is:

- A) Zero Trust**
- B) Defense in depth**
- C) Fail-open**
- D) Security through obscurity**

4. Corporate scenario (Network Infrastructure Concepts): During a security review and compliance audit aligned with the CompTIA Security+ exam (SY0-701, Section 3.1), an analyst evaluates the following technical requirement: 'SDN: centralized control plane; programmable and dynamic rules'. What is the best recommendation or applicable control classification?

- A) Temporarily disable audit logs to accelerate system throughput**
- B) Implement the corresponding solution: SDN: centralized control plane; programmable and dynamic rules...**
- C) Replace preventive technical controls with informal terms of adherence**
- D) Allow shared access among administrators using generic credentials**

5. Corporate scenario (Network Infrastructure Concepts): During a security review and compliance audit aligned with the CompTIA Security+ exam (SY0-701, Section 3.1), an analyst evaluates the following technical requirement: 'Microsegmentation: isolates each workload — Zero Trust applied to the network'. What is the best recommendation or applicable control classification?

- A) Temporarily disable audit logs to accelerate system throughput**
- B) Implement the corresponding solution: Microsegmentation: isolates each workload — Zero Trust applied to the network.**
- C) Replace preventive technical controls with informal terms of adherence**
- D) Allow shared access among administrators using generic credentials**

6. Corporate scenario (Network Infrastructure Concepts): During a security review and compliance audit aligned with the CompTIA Security+ exam (SY0-701, Section 3.1), an analyst evaluates the following technical requirement: 'VPC: your network isolated from the rest of the provider within the cloud.' What is the best recommendation or applicable control classification?

- A)** Implement the corresponding solution: VPC: your network isolated from the rest of the provider within the cloud...
- B)** Temporarily disable audit logs to accelerate system throughput
- C)** Replace preventive technical controls with informal terms of adherence
- D)** Allow shared access among administrators using generic credentials

7. Corporate scenario (Network Infrastructure Concepts): During a security review and compliance audit aligned with the CompTIA Security+ exam (SY0-701, Section 3.1), an analyst evaluates the following technical requirement: 'centralized control plane managing network routing' = SDN. What is the best recommendation or applicable control classification?

- A)** Temporarily disable audit logs to accelerate system throughput
- B)** Replace preventive technical controls with informal terms of adherence
- C)** Implement the corresponding solution: 'centralized control plane managing network routing' = SDN...
- D)** Allow shared access among administrators with generic credentials

8. Corporate scenario (Network Infrastructure Concepts): During a security review and compliance audit aligned with the CompTIA Security+ exam (SY0-701, Section 3.1), an analyst evaluates the following technical requirement: 'isolates individual workloads' = microsegmentation.' What is the best recommendation or applicable control classification?

- A)** Temporarily disable audit logs to accelerate system throughput
- B)** Replace preventive technical controls with informal terms of adherence
- C)** Implement the corresponding solution: 'isolates individual workloads' = microsegmentation...
- D)** Allow shared access among administrators using generic credentials

9. Corporate scenario (Network Infrastructure Concepts): During a security review and compliance audit aligned with the CompTIA Security+ exam (SY0-701, Section 3.1), an analyst evaluates the following technical requirement: 'Scenario: 'cloud network configured so that the test workload cannot talk to the production workload' = microsegmentation.' What is the best recommendation or applicable control classification?

- A)** Implement the corresponding solution: Scenario: 'cloud network configured so that the test workload cannot talk to ...
- B)** Temporarily disable audit logs to accelerate system throughput
- C)** Replace preventive technical controls with informal terms of adherence
- D)** Allow shared access among administrators using generic credentials

10. Corporate scenario (Network Infrastructure Concepts): During a security review and compliance audit aligned with the CompTIA Security+ exam (SY0-701, Section 3.1), an analyst evaluates the following technical requirement: 'SDN: centralized control plane; programmable and dynamic rules'. What is the best recommendation or applicable control classification?

- A)** Temporarily disable audit logs to speed up system throughput
- B)** Replace preventive technical controls with informal terms of adherence
- C)** Allow shared access among administrators with generic credentials
- D)** Implement the corresponding solution: SDN: centralized control plane; programmable and dynamic rules...

COMPREHENSIVE ANSWER KEY & DISTRACTOR BREAKDOWN

1) A — VPC (Virtual Private Cloud).

✓ **Why it's correct:** A VPC (Virtual Private Cloud) is a logically isolated virtual network within a public cloud provider, where the customer defines the IP address space (CIDR), public and private subnets, route tables, internet/NAT gateways, access control lists (ACLs), and security groups. According to CompTIA Security+ (domain 3.4), VPC is the solution that enables network segmentation, resource isolation, and control of east-west and north-south traffic, exactly matching the statement: 'your isolated private network within the cloud, where you define subnets and routing'.

✗ **Why the others are wrong:**

- *CDN:* A CDN (Content Delivery Network) is a geographically distributed network of servers that cache copies of content for fast delivery to end users. It does not provide network isolation, subnet definition, or private routing; its function is caching and content acceleration. Therefore, it is not the correct answer.
- *WAF:* A WAF (Web Application Firewall) is a firewall specialized in the application layer (HTTP/HTTPS) that inspects and filters web traffic to protect against attacks such as SQL injection and XSS. It acts on application protection, not on creating isolated private networks with subnets and routing. Thus, it is incorrect for the context.
- *NAT gateway:* A NAT gateway is a network component that allows instances in private subnets to access the internet without exposing their private IP addresses, performing address translation. It is an element within a VPC, but it is not the isolated private network itself; it does not define subnets or global routing. Therefore, it does not meet the statement.

2) A — Virtual Private Cloud - isolated network in the cloud.

✓ **Why it's correct:** VPC stands for Virtual Private Cloud. It is a logically isolated virtual network within a public cloud infrastructure, such as AWS, Azure, or GCP. A VPC allows the customer to define and control network aspects such as IP address ranges, subnets, route tables, gateways, and security groups, creating a segmented and secure environment that simulates a traditional private network. Its primary function is to provide network isolation, traffic control, and secure connectivity between cloud resources and on-premises networks (via VPN or Direct Connect). In CompTIA Security+, VPC is an example of segmentation and isolation control in cloud environments, essential for implementing defense in depth and reducing the attack surface.

✗ **Why the others are wrong:**

- *Virtual Public Cloud - shared network:* There is no standard term 'Virtual Public Cloud' as a concept in cloud security. Public clouds are multi-tenant environments where resources are shared, but VPC specifically creates a virtual private network within that public cloud, isolating the customer's traffic. This option is wrong because it reverses the meaning: VPC is not a shared public network, but a logically isolated private network within a public infrastructure.
- *Virtual Computing - elastic processing:* Virtual Computing refers to the abstraction of computing resources, such as virtual machines and containers, not an isolated network. Elastic processing is the ability to automatically scale resources based on demand. Neither concept corresponds to the acronym VPC or its network function. This option is incorrect because it describes a computing model, not a virtual private network.
- *Volume Persistent Cloud - storage:* Volume Persistent Cloud is not a recognized term in cloud architecture. Persistent cloud storage is typically called block storage (e.g., Amazon EBS) or object storage (e.g., Amazon S3). VPC is not a storage service, but a network service. This option is wrong

because it confuses the acronym and function of VPC with a storage concept that does not exist under that nomenclature.

3) B — Defense in depth.

✓ **Why it is correct:** The correct answer is 'Defense in depth'. Defense in depth is a security strategy that employs multiple layers of security controls, so that if one layer fails, others still protect the system. This includes physical, technical, and administrative controls, such as firewalls, antivirus, multi-factor authentication (MFA), user training, policies, encryption, among others. The goal is not to rely on a single security measure. CompTIA Security+ SY0-701, in the Architecture and Design domain (2.1), describes defense in depth as a core concept. The question describes exactly this: using multiple layers of defense so as not to depend on just one.

✗ **Why the others are wrong:**

- *Zero Trust*: Zero Trust is a security model that assumes no entity, inside or outside the network, is trusted by default. It requires continuous verification of identity and context for each access, following the principle of 'never trust, always verify'. Although Zero Trust may include multiple layers, its main focus is explicit verification and least privilege, not simply the idea of defense layers. The question describes the strategy of multiple layers, which is defense in depth, not the Zero Trust model.
- *Fail-open*: Fail-open is a failure mode in which, when a security system fails, it allows traffic or access to pass, rather than blocking it. This is the opposite of a secure posture, which generally prefers fail-secure (fail-closed). Fail-open is not a layered defense strategy; it is a failure behavior that can be configured on some devices, but does not describe the practice of using multiple defenses.
- *Security through obscurity*: Security through obscurity is the practice of hiding implementation details or information as a form of protection, for example, using non-standard ports or secret file names. It is considered a weak practice, as it does not resist a determined attacker who discovers the obscurity. It does not refer to multiple layers of defense; on the contrary, it is a single fragile layer. The question describes the combination of various defenses, which is defense in depth.

4) B — Implement the corresponding solution: SDN: centralized control plane; programmable and dynamic rules....

✓ **Why it is correct:** According to the official CompTIA Security+ (SY0-701) exam objectives for lesson 3.1 (Network Infrastructure Concepts): SDN: centralized control plane; programmable and dynamic rules. SDN separates the control plane from the data plane, enabling centralized management and programmability for dynamic network configurations.

✗ **Why the others are wrong:**

- *Temporarily disable audit logs to accelerate system throughput*: Directly violates the Accounting pillar and forensic traceability. Without logs, the team is blind to detecting incidents and violates compliance standards (PCI DSS, ISO 27001).
- *Replace preventive technical controls with informal terms of adherence*: Informal terms are administrative controls, rely solely on user behavior, and do not block automated attacks, injections, or malware at runtime.
- *Allow shared access among administrators using generic credentials*: Destroys individual accountability and non-repudiation. Makes it impossible to prove who performed critical changes or malicious commands.

5) B — Implement the corresponding solution: Microsegmentation: isolates each workload — Zero Trust applied to the network..

✓ **Why it is correct:** According to the official CompTIA Security+ (SY0-701) exam objectives for

lesson 3.1 (Network Infrastructure Concepts): Microsegmentation: isolates each workload — Zero Trust applied to the network. Microsegmentation divides the network into logical segments down to individual workloads, enforcing granular security policies and preventing lateral movement. It is a key implementation of Zero Trust principles in network infrastructure.

✗ Why the others are wrong:

- *Temporarily disable audit logs to accelerate system throughput:* Directly violates the Accounting pillar and forensic traceability. Without logs, the team is blind to incident detection and breaks compliance standards (PCI DSS, ISO 27001).
- *Replace preventive technical controls with informal terms of adherence:* Informal terms are administrative controls that rely solely on user behavior and do not block automated attacks, injections, or runtime malware.
- *Allow shared access among administrators using generic credentials:* Destroys individual accountability and non-repudiation. It makes it impossible to prove who performed critical changes or malicious commands.

6) A — Implement the corresponding solution: VPC: your network isolated from the rest of the provider within the cloud....

✓ **Why it is correct:** According to the official CompTIA Security+ (SY0-701) exam objectives for lesson 3.1 (Network Infrastructure Concepts): VPC: your network isolated from the rest of the provider within the cloud. A Virtual Private Cloud (VPC) is a logically isolated section of a cloud provider's infrastructure where you can launch resources in a virtual network that you define, providing network isolation and security.

✗ Why the others are wrong:

- *Temporarily disable audit logs to accelerate system throughput:* Directly violates the Accounting pillar and forensic traceability. Without logs, the team is blind to detecting incidents and breaks compliance standards (PCI DSS, ISO 27001).
- *Replace preventive technical controls with informal terms of adherence:* Informal terms are administrative controls that rely solely on user conduct and do not block automated attacks, injections, or malware at runtime.
- *Allow shared access among administrators using generic credentials:* Destroys individual accountability and non-repudiation. It makes it impossible to prove who performed critical changes or malicious commands.

7) C — Implement the corresponding solution: 'centralized control plane managing network routing' = SDN....

✓ **Why it's correct:** According to the official CompTIA Security+ (SY0-701) exam objectives for lesson 3.1 (Network Infrastructure Concepts): 'centralized control plane managing network routing' = SDN.

✗ Why the others are wrong:

- *Temporarily disable audit logs to accelerate system throughput:* Directly violates the Accounting pillar and forensic traceability. Without logs, the team is blind to detecting incidents and breaks compliance standards (PCI DSS, ISO 27001).
- *Replace preventive technical controls with informal terms of adherence:* Informal terms are administrative controls, rely solely on user conduct, and do not block automated attacks, injections, or malware at runtime.
- *Allow shared access among administrators with generic credentials:* Destroys individual accountability and non-repudiation. Makes it impossible to prove who performed critical changes or malicious commands.

8) C — Implement the corresponding solution: 'isolates individual workloads' = microsegmentation....

✓ **Why it's correct:** According to the official CompTIA Security+ (SY0-701) exam objectives for lesson 3.1 (Network Infrastructure Concepts): 'isolates individual workloads' = microsegmentation. Microsegmentation is a security technique that creates secure zones in data centers and cloud deployments, isolating individual workloads from one another to reduce the attack surface and limit lateral movement.

✗ **Why the others are wrong:**

- *Temporarily disable audit logs to accelerate system throughput:* Directly violates the Accounting pillar and forensic traceability. Without logs, the team is blind to detect incidents and breaks compliance standards (PCI DSS, ISO 27001).
- *Replace preventive technical controls with informal terms of adherence:* Informal terms are administrative controls, rely solely on user conduct, and do not block automated attacks, injections, or malware at runtime.
- *Allow shared access among administrators using generic credentials:* Destroys individual accountability and non-repudiation. Makes it impossible to prove who made critical changes or malicious commands.

9) A — Implement the corresponding solution: Scenario: 'cloud network configured so that the test workload cannot talk to

✓ **Why it is correct:** According to the official CompTIA Security+ (SY0-701) exam objectives for lesson 3.1 (Network Infrastructure Concepts): Scenario: 'cloud network configured so that the test workload cannot talk to the production workload' = microsegmentation. Microsegmentation is a security technique that creates fine-grained security zones within a data center or cloud environment, isolating workloads from each other. The correct action is to implement the corresponding microsegmentation controls.

✗ **Why the others are wrong:**

- *Temporarily disable audit logs to accelerate system throughput:* Directly violates the Accounting pillar and forensic traceability. Without logs, the team is blind to detecting incidents and breaks compliance standards (PCI DSS, ISO 27001).
- *Replace preventive technical controls with informal terms of adherence:* Informal terms are administrative controls that rely solely on user behavior and do not block automated attacks, injections, or malware at runtime.
- *Allow shared access among administrators using generic credentials:* Destroys individual accountability and non-repudiation. It makes it impossible to prove who performed critical changes or malicious commands.

10) D — Implement the corresponding solution: SDN: centralized control plane; programmable and dynamic rules....

✓ **Why it's correct:** According to the official CompTIA Security+ (SY0-701) exam objectives for lesson 3.1 (Network Infrastructure Concepts): SDN: centralized control plane; programmable and dynamic rules.

✗ **Why the others are wrong:**

- *Temporarily disable audit logs to speed up system throughput:* Directly violates the Accounting pillar and forensic traceability. Without logs, the team is blind to detecting incidents and breaks compliance standards (PCI DSS, ISO 27001).
- *Replace preventive technical controls with informal terms of adherence:* Informal terms are administrative controls, rely solely on user conduct, and do not block automated attacks,

injections, or malware at runtime.

- *Allow shared access among administrators with generic credentials:* Destroys individual accountability and non-repudiation. Makes it impossible to prove who performed critical changes or malicious commands.